

移动开发第九周

Navigate between screens with Compose

English



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Note: The `composable()` function is an extension function of `NavGraphBuilder`.

1. Call the `composable()` function, passing in `CupcakeScreen.Start.name` for the `route`.

```
import androidx.navigation.compose.composable

NavHost(
    navController = navController,
    startDestination = CupcakeScreen.Start.name,
    modifier = Modifier.padding(innerPadding)
) {
    composable(route = CupcakeScreen.Start.name) {
    }
}
```

2. Within the trailing lambda, call the `StartOrderScreen` composable, passing in `quantityOptions` for the `quantityOptions` property. For the `modifier` pass in `Modifier.fillMaxSize().padding(dimensionResource(R.dimen.padding_medium))`

```
import androidx.compose.foundation.layout.fillMaxSize
import androidx.compose.ui.res.dimensionResource
import com.example.cupcake.ui.StartOrderScreen
```

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5. Call the `SelectOptionScreen` composable.

```
composable(route = CupcakeScreen.Flavor.name) {
    val context = LocalContext.current
    SelectOptionScreen()
}
```

6. The flavor screen needs to display and update the subtotal when the user selects a flavor. Pass in `uiState.price` for the `subtotal` parameter.

```
composable(route = CupcakeScreen.Flavor.name) {
    val context = LocalContext.current
    SelectOptionScreen(
        subtotal = uiState.price
    )
}
```

7. The flavor screen gets the list of flavors from the app's string resources. Transform the list of resource IDs into a list of strings using the `map()` function and calling `context.resources.getString(id)` for each flavor.

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3. Now that the `StartOrderScreen` composable expects a value for `onNextButtonClicked`, find the `StartOrderPreview` and pass an empty lambda body to the `onNextButtonClicked` parameter.

```
@Preview
@Composable
fun StartOrderPreview() {
    CupcakeTheme {
        StartOrderScreen(
            quantityOptions = DataSource.quantityOptions,
            onNextButtonClicked = {},
            modifier = Modifier
                .fillMaxSize()
                .padding(dimensionResource(R.dimen.padding_medium))
        )
    }
}
```

2. Below the `onCancelButtonClicked` parameter, add another parameter of type `() -> Unit` named `onNextButtonClicked` with a default value of `{}`.

```
@Composable
fun SelectOptionScreen(
    subtotal: String,
    options: List<String>,
    onSelectionChanged: (String) -> Unit = {},
    onCancelButtonClicked: () -> Unit = {},
    onNextButtonClicked: () -> Unit = {},
    modifier: Modifier = Modifier
)
```

3. Pass in `onCancelButtonClicked` for the cancel button's `onClick` parameter.

```
OutlinedButton(
    modifier = Modifier.weight(1f),
    onClick = onCancelButtonClicked
) {
    Text(stringResource(R.string.cancel))
}
```

3. The `OrderSummaryScreen` composable now expects values for `onSendButtonClicked` and `onCancelButtonClicked`. Find the `OrderSummaryPreview`, pass an empty lambda body with two `String` parameters to `onSendButtonClicked` and an empty lambda body to the `onCancelButtonClicked` parameters.

```
@Preview
@Composable
fun OrderSummaryPreview() {
    CupcakeTheme {
        OrderSummaryScreen(
            orderUiState = OrderUiState(0, "Test", "Test", "$300.00"),
            onSendButtonClicked = { subject: String, summary: String -> },
            onCancelButtonClicked = {},
            modifier = Modifier.fillMaxHeight()
        )
    }
}
```

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10. Within the lambda passed into `startActivity()`, create an activity from the Intent by calling the class method `createChooser()`. Pass intent for the first argument and the `new_cupcake_order` string resource.

```
context.startActivity(
    Intent.createChooser(
        intent,
        context.getString(R.string.new_cupcake_order)
    )
)
```

11. In the `CupcakeApp` composable, in the call to `composable()` for the `CupcakeScreen.Summary.name`, get a reference to the context object so that you can pass it to the `shareOrder()` function.

```
composable(route = CupcakeScreen.Summary.name) {
    val context = LocalContext.current
    ...
}
```

12. In the lambda body of `onSendButtonClicked()`, call `shareOrder()`, passing in the `context`, `subject`, and

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8. Get the solution code

To download the code for the finished codelab, you can use these git commands:

```
$ git clone https://github.com/google-developer-training/basic-android-kotlin-compose-training-cupcake
$ cd basic-android-kotlin-compose-training-cupcake
$ git checkout navigation
```

Alternatively, you can download the repository as a zip file, unzip it, and open it in Android Studio.

[Download zip](#)

Note: The solution code is in the `navigation` branch of the downloaded repository.

If you want to see the solution code for this codelab, view it on [GitHub](#).



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8. Summary

Congratulations! You've learned how to test the Jetpack Navigation component. You also learned some fundamental skills for writing UI tests, such as writing reusable helper methods, how to leverage `setContent()` to write concise tests, how to set up your tests with the `@Before` annotation, and how to think about maximum test coverage. As you continue to build Android apps, remember to keep writing tests alongside your feature code!